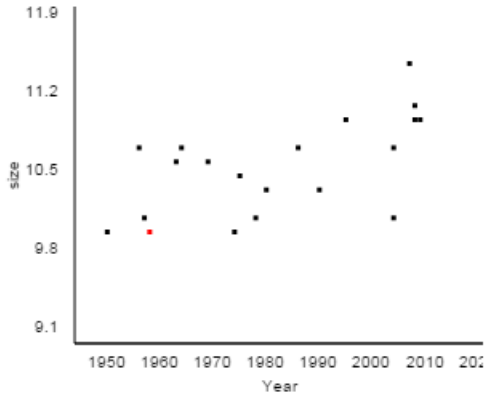
Chapter 10

Solutions to exploration 10.1

1. It appears that the plate on the left has more food on it.
2. Explanatory: year of manufacture, quantitative
3. Response: plate size, quantitative
4. An association would mean that plate sizes changing over time, for example they could be trending upward with year.
5. The observational units are dinner plates sold on eBay.com March 30, 2010, there are 20 plates.
6. This study is an observational study because the investigators didn't manipulate any of the variables, rather they simply observed them.
7. Year is on the x-axis, size is on the y-axis
8. Red dot is 1958, 10



9. Positive, as year increases, so does size of plate (on average)
10. Linear
11. Fairly strong
12. There do not seem to be any observational units (dots) that fall outside the overall pattern.
13. Positive, because there was a positive association between the variables
14. Answers will vary, 0.65
15. 0.604
16. Seems to be a high plate size for a plate in 1950. The correlation moved closer to zero (0.371)

Solutions to exploration 10.2

1. 0
2. a. No
   b. Responses will vary, should all be close to 0 though
   c. Yes
3. a. Responses will vary to correspond with responder's birthday
   b. depends on answer to part a
4. a. answers will depend on birth month
   b. median draft numbers by month

Month	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Median draft number	211	210	256	225	226	207.5	188	145	168	201	131.5	100

c. Median draft numbers trend downward as months increase from Jan to Dec.

5. A negative association. Doesn't seem consistent with the scatter plot until you notice that there does tend to be fewer "dots" in the upper right and lower left corners.
6. a. The population correlation is zero and we just got this result by random chance OR There really is an association between the draft lottery number and the sequential date.
   b. i. The statistic is the correlation coefficient. ii. Simulate by shuffling the draft lottery numbers and reassign to the sequential dates. Do this many times, recalculating the correlation coefficient each time to see what values we would expect for the correlation coefficient if this truly were governed by random chance. iii. Compare the observed correlation coefficient to the distribution of simulated correlation coefficients to see whether it is a typical valueor a value that is unlikely to have occurred assuming the null to be true.
7. a. After one shuffle, answers will vary.
   b. Answers will vary, for example: 0.002, 0.039, -0.027, 0.09, -0.039
   c. No
   d. Center is 0. If the null hypothesis is true, that means there is no underlying association between the two variables, so we would expect correlations around zero.
   e. Approximate p-value is 0
   f. The probability of getting a correlation coefficient of -0.226 or more extreme (in either direction) assuming the null hypothesis of no association is true.
   g. The p-value of zero offers strong evidence against the null hypothesis and in support of the alternative hypothesis that there is an association between draft lottery number and sequential birth date. In all the shufflings and recalculations of simulated correlation coefficients, not one was as extreme as -0.226 from the study.
8. Answers will vary, should be a p-value around 0.80. It appears this lottery was conducted by a fair, random process because the observed correlation coefficient is a typical correlation coefficient we'd expect from a random process. The null distribution generated from the 1971 data would be essentially the same as the null distribution from the 1970 data (just lose one day).

Solutions to Exploration 10.3

1. a. observational units are the 20 statistics students
   b. explanatory variable is foot length, quantitative; response variable is height, quantitative
   c. This is an observational study
2. The scatterplot shows a moderately strong positive linear association between foot length and height.
3. Yes

4. a. Answers will vary, for example, movable line was: predicted height = 26.27 + 1.46 (footlength)
   b. Because there weren't many points far away from the line.
   c. No
   d. The line that is closest to all the points
5. Above: positive residual Below: negative residual
6. Answers will vary, e.g., SAE = 63.98
7. a. Answers will vary, e.g., SSE = 275.22
   b. Answers will vary, e.g., new equation: predicted height = 32.1 + 1.15 (footlength); new SSE = 235.01
8. a. predicted height = 38.30 + 1.03 (footlength)
   b. Yes
   c. The slope is positive so that tells us there is a positive association between the two variables (as one increases the other one does as well).
9. a. 67.14
   b. 68.17
   c. differ by 1.03, this is the slope.
   d. The slope of the regression line predicting height based on foot length is 1.03, meaning that for every additional one cm increase in footlength, the predicted height increases by 1.03 inches
10. a. 38.30 inches
    b. This is the y-intercept.
    c. There are no individuals with a foot length of 0 in the data. The lowest footlength is 22 cm., and most range from the mid 20s to the mid 30s.
11. a. 74-71.26=2.74 b. Above the line, because the residual is positive.
12. a. Answers will vary, the regression line changes a little, but not much
    b. Moving the very small foot size or the very large one seem to have the most influence on the line. The slope changes the most when the small value is moved up, or the large value is moved down.
13. a. SSE =475.75
    b. 0
    c. No
14. SSE is smaller from the regression line. This makes sense because the regression line is the best fitting line and does so by minimizing the sum of the squared errors.
15. a. R^2 =50.6%
    b. 50.6%, this means that 50.6% of the variation in people's height is attributed to changes in their foot length.
    c. $r = 0.711$, and $r^2 = (0.711^2) \approx .506$, the coefficient of determination.

Solutions to Exploration 10.4

1. H_0: There is no association between hand width and estimated weight of bean bag
   H_a: There is an association between hand width and estimated weight of bean bag

2. There appears to be a weak negative linear association.
3. Slope = -3.76. For each 1 cm increase in hand width, the predicted perceived weight of beanbag decreases 3.76 grams.
4. a. -3.76,
   b. shuffle the 46 slips of paper with perceived weights on them and deal one to each of the slips with hand width written on it. Compute the least squares regression line and record the slope. Repeat this process several times, like 1000, to get a good idea what simulated slopes to expect if there is no association between one's hand width and perception of weight of a beanbag.
   c. Now compare the slope from the least squares regression line fit to the original data to see whether it is a typical value unlikely to have occurred assuming no association.
5. a. Answers will vary, example response: 5 simulated slopes were 2.02, 1.22, 1.85, -2.59, 0.65. None were as extreme as the slope from the least squares regression line for the original data (-3.76).
   b. Answers will vary, typically around 0.020.
   c. There are a few slopes smaller than that of the one observed, but a very small proportion.
   d. The p-value is the probability of getting a slope from a least squares regression line that is -3.76 or more extreme assuming there is no association between hand width and estimated beanbag weight.
   e. Yes
6. The p-value is the same.
7. a. We would need to know more about our sample before generalizing. Were they randomly sampled from a larger population? What were their ages? Were they all females?
   b. No, there was no random assignment to balance confounding variables. In this particular case, hand width is a variable that can't be randomly assigned, only observed.

Solutions to Exploration 10.5

1. H_0: There is no association between brain density and Facebook friends in the population
   H_a: There is an association between brain density and Facebook friends in the population
2. The scatterplot shows a moderate, positive linear association between the number of Facebook friends and brain density. There do not appear to be any unusual observations.
3. 0.2009 The sign is positive on slope indicating a positive association.
4. a. Take 80 slips of paper. Write the brain densities on 40 and the number of Facebook friends (in hundreds) on the other 40. Shuffle the 40 responses (brain densities) and deal one to each of the 40 Facebook friend slips. The null hypothesis of no association is modeled by shuffling the response variable because that breaks the association between the original pairs and results in a slope based on data with no association.
   b. Answers will vary,e.g., 5 simulated slopes are: 0.179, -0.1387, -.0315, 0.0525, 0.1032. None are as large or larger than our sample slope of 0.2009.
   c. Answers will vary, e.g., p-value = 0.009

d. 0.009 is the probability that we would have observed a least squares regression line with slope of 0.2009 or more extreme if there were no association between brain density and Facebook friends.

e. Yes, for the population of university students in London.

5. Bell-shaped and symmetric
6. 0.366
7. $t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}} = \frac{0.366}{\sqrt{\frac{1-0.366^2}{40-2}}} = \frac{0.366}{\sqrt{\frac{0.866}{38}}} = 2.42$
8. Yes
9. Yes
10. Yes
11. Yes, the validity conditions were met.
12. Theory-based p-value = 0.0204, standardized *t* statistic = 2.42. Both the p-value and the standardized statistic offer strong evidence against the null and in favor of the alternative that there is an association between brain density and Facebook friends.
13. Null: Population slope (β) is 0, Alt: Population Slope (β) is not 0
14. Null: Population correlation (ρ) is 0, Alt: Population correlation (ρ) is not 0.
15. (0.0329 , 0.3689) We are 95% confident that in the population of interest, brain density increases (on average) between 0.0329 and 0.3689 for each additional unit (100 Facebook friends) increase of number of Facebook friends. Because the sample consisted of university students in London, that may be the population to which we generalize our results
16. a. 0.023
    b. 0
    c. Validity conditions are not met for the theory-based test.
17. a. Can probably generalize to university students in London as long as those who volunteered are not any different from those who didn't.

    b. We cannot make any cause-and-effect conclusions because there was no random assignment of the explanatory variable, number of Facebook friends.

    c. We cannot make causal conclusions in either direction because this was an observational study and there are many potential confounding variables.
18. A false positive would mean that you found a significant association when really there was no association. When multiple tests are done on the same data to search for significant associations, the false positive rate increases with each additional test.